SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – MCQ Test

Course Code:	CSA0305	**Course Title:**	Data Structures
CO Assessed:	CO3 – Precision (BL3, BL5)	**Assessment:**	Assessment Tool 3 – MCQ Test
Weightage:	30% of CIA	**Total Marks:**	100
CO3 Statement:	*Solve problems for optimized data storage and retrieval using Binary Search Trees, AVL Trees, B-Trees, Red-Black Trees, and Splay Trees.*		
Student Name:	K.Karthik	**Reg. No.:**	192525040
Section / Batch:	AI & ML	**Date:**	12/08/2026

Code of Conduct

I K.Karthik / 192525040 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: K.Karthik

Instructions

- This test contains 25 MCQ Test questions (4 marks each), Total: 100 Marks.
- When a student answer using MCQ, in evaluation the answer should be with following requirements: Identify the most appropriate answer, conceptual understanding, problem-solving ability, application of knowledge.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Preliminaries – Tree Terminology	Root, height, level, siblings and sub tree	1–2	8
Binary Tree, Binary Search Tree	Properties, binary tree and binary search tree, insertion and deletion	3–9	28
Tree Traversals	Traversal types, In order, Post order and Pre order	10–14	20
AVL Tree	Balancing factor, Rotation, Types of rotation, examples	15–16	8
Applications of Search Trees	Usage of binary tree	17	4
B Tree - 2-3 tree, 2-3-4 tree	Properties, Example and Operations	18–19	8
TRIE	Properties, structures and Operations	20	4
Red Black Tree	Properties, Balancing and Operations	21–22	8
Splay Tree	Splaying, Types and examples for insertion and deletion	23–25	12
		GRAND TOTAL:	**100 Marks**

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KONIJETI KARTHIK
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CO3_AT3_MCQ

QUESTIONS

Q1
Tree Traversal
4 marks

Q2
Binary Search Tree
4 marks

Q3
Red-Black Tree
4 marks

Q4
Red-Black Tree
4 marks

Q5
B-Tree

Q1 **Tree Traversal** 4 marks

You analyze the post-order traversal of a tree representing project tasks: [D, E, B, F, G, C, A]. You want to know

A. 3

B. 4

C. 5

D. 2

Save Answer

Next →

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QUESTIONS

Q1
Tree Traversal
4 marks

Q2
Binary Search Tree
4 marks

Q3
Red-Black Tree
4 marks

Q4
Red-Black Tree
4 marks

Q5
B-Tree
4 marks

Q25 **Splay Tree**

4 marks

After accessing a node in a Splay Tree, that node is moved to:

A. Leaf

B. Middle

C. Root

D. NULL

Save Answer

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CO2_AT2_DEBUGGING & OPTIMIZATION TASK (Set 1) OPEN ✓ Completed

DEBUGGING & OPTIMIZATION TASK

10/10 answered

Review →

CO3_AT3_MCQ OPEN ✓ Completed

Multiple Choice Questions - Non Linear Data Structures

25/25 answered

Review →

CO1_AT1_C CODE OUTPUT PREDICTION OPEN ✓ Completed

C CODE OUTPUT PREDICTION

10/10 answered

Review →